

AI Fundamentals

Ten days on flat, solid ground: what AI actually is, how it learns, and where it breaks. The first waypoints on a 30-day field journal.

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DAY 01 / 30

BASECAMP · FUNDAMENTALS

What is AI, really?

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Everyone says "AI" like it's one thing. It isn't.

- ◆ Software that finds patterns in data instead of following hand-written rules
- ◆ A spam filter and a chatbot are both "AI" — just different points on a spectrum
- ◆ Ask what pattern it learned, and from what data, not "is this magic"

DAY 01 / 30

“If you could finally understand one thing about AI, what would it be? Drop it in the comments — I'll make sure it's covered somewhere in this series.”

Follow along — Day 2 drops tomorrow.

#AI

#MachineLearning

#LearnInPublic

#ArtificialIntelligence

DAY 02 / 30

BASECAMP · FUNDAMENTALS

AI, ML, Deep Learning, GenAI: the family tree

AI, ML, Deep Learning, GenAI: the family tree

I used to use these four terms interchangeably. That was a mistake — they're nested inside each other, like Russian dolls.

- ◆ Artificial Intelligence (AI) is the broadest term: any system that mimics intelligent behavior.
- ◆ Machine Learning (ML) is a subset of AI: systems that improve at a task by learning from data, rather than being explicitly programmed.
- ◆ Deep Learning is a subset of ML: it uses layered neural networks (more on those in a few days) to learn especially complex patterns — the technique behind image recognition and speech-to-text.
- ◆ Generative AI is a subset of deep learning: models trained to produce new content — text, images, audio — rather than just classify or predict a number.

DAY 02 / 30

“Which of these four terms do you think gets misused the most in your industry?”

Follow along – Day 3 drops tomorrow.

#AI

#MachineLearning

#DeepLearning

#GenerativeAI

DAY 03 / 30

BASECAMP · FUNDAMENTALS

How machines "learn": training data explained simply

How machines "learn": training data explained simply

The word "learn" makes it sound like the computer is thinking. It's really just adjusting numbers until it stops being wrong as often.

- ◆ The model guesses, gets corrected, adjusts its internal numbers — repeat billions of times
- ◆ No understanding required, just automated trial and error at massive scale
- ◆ Data quality sets the ceiling: garbage in, garbage out

“Next time you evaluate an AI product, ask what data it was trained on before you ask what algorithm it uses. The data question matters more.”

Follow along – Day 4 drops tomorrow.

#MachineLearning

#DataScience

#AI

#TrainingData

DAY 04 / 30

BASECAMP · FUNDAMENTALS

Supervised vs unsupervised vs reinforcement learning

Supervised vs unsupervised vs reinforcement learning

There are three main ways to teach a machine, and they map surprisingly well onto how humans learn too.

- ◆ Supervised learning: learning from labeled examples with a known right answer. Like flashcards — "here's the input, here's the correct output, learn the pattern." Used for spam detection, price prediction, medical image diagnosis.
- ◆ Unsupervised learning: finding structure in data with no right answers given. Like being handed a messy pile of photos and asked to sort them into groups without being told what the groups are. Used for customer segmentation, anomaly detection.
- ◆ Reinforcement learning: learning through trial, error, and reward — no labeled data at all, just a signal for "good" or "bad" outcomes. Like training a dog with treats. Used for game-playing AI, robotics, and — this part surprises people — a key step in training chatbots to be helpful (RLHF: [Reinforcement Learning from Human Feedback](#)).

DAY 04 / 30

“Which of these three feels most intuitive to you, and which still feels like a black box?”

Follow along – Day 5 drops tomorrow.

#MachineLearning

#ReinforcementLearning

#AI

#DataScience

DAY 05 / 30

BASECAMP · FUNDAMENTALS

Neural networks in plain English

Neural networks in plain English

Neural networks sound intimidating because of the name. The actual idea is simpler than most spreadsheets I've built.

- ◆ Layers of tiny weighted decision points passing signals forward
- ◆ Early layers learn simple patterns, later layers learn abstract ones
- ◆ "Neural" is a loose metaphor, not a brain simulation — it's weighted math

DAY 05 / 30

“Does knowing it's "just" weighted math make AI feel less magical, or more impressive?”

Follow along – Day 6 drops tomorrow.

#NeuralNetworks

#DeepLearning

#AI

#MachineLearning

DAY 06 / 30

BASECAMP • FUNDAMENTALS

What is a Large Language Model (LLM)?

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An LLM — the tech behind ChatGPT, Claude, Gemini — is, at its core, a very sophisticated next-word predictor.

- ◆ At its core: a very sophisticated next-word predictor
- ◆ "Large" means huge training data plus billions of parameters
- ◆ No fact database — it generates plausible text, usually (not always) true

“Understanding "next-token prediction" is the single most useful mental model for using these tools well: you're not talking to a database, you're steering a very good pattern-completion engine.”

Follow along – Day 7 drops tomorrow.

#LLM

#GenerativeAI

#ChatGPT

#AI

DAY 07 / 30

BASECAMP · FUNDAMENTALS

Training vs inference: why ChatGPT doesn't "learn" from your chats

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A question I get a lot: "If I correct the AI in a conversation, does it remember that for next time?" Short answer: no. And understanding why clears up a lot of confusion about how these systems actually work.

- ◆ Training: the expensive, offline process (Day 3) where the model's weights get adjusted across a massive dataset. This happens once, over weeks, on huge clusters of specialized hardware, before you ever see the model.
- ◆ Inference: what happens every time you send a message. The model's weights are now frozen. It's just running the "arithmetic" from Day 5 forward — reading your input and generating output. No weights change during inference, ever.

DAY 07 / 30

“Did you assume AI chatbots were continuously learning from your conversations? You're not alone — most people do.”

Follow along – Day 8 drops tomorrow.

#AI

#LLM

#MachineLearning

#GenerativeAI

DAY 08 / 30

BASECAMP · FUNDAMENTALS

Tokens & embeddings: how AI understands language

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AI models don't read words. They read numbers. Here's how text becomes math.

- ◆ Tokenization chops text into word and sub-word chunks the model can process
- ◆ Embeddings turn tokens into numbers that capture meaning as geometry
- ◆ Similar meanings end up mathematically close together

DAY 08 / 30

“Once you see language as coordinates in space, a lot of "how does AI even understand me" stops feeling mysterious and starts feeling like math.”

Follow along – Day 9 drops tomorrow.

#AI

#NLP

#MachineLearning

#Embeddings

DAY 09 / 30

BASECAMP • FUNDAMENTALS

Hallucinations: why AI confidently makes things up

Hallucinations: why AI confidently makes things up

The single most important thing to understand before you trust AI output for anything important: it was never optimized to tell the truth. It was optimized to sound plausible.

- ◆ Asking for specific citations, statistics, or quotes
- ◆ Questions about very recent events (past the training cutoff)
- ◆ Obscure or niche topics with little training data
- ◆ Leading questions that assume a false premise

DAY 09 / 30

“What's the most confidently wrong answer AI has ever given you?”

Follow along – Day 10 drops tomorrow.

#AI

#Hallucination

#LLM

#ResponsibleAI

DAY 10 / 30

BASECAMP · FUNDAMENTALS

AI ethics & bias: the human side of the machine

AI ethics & bias: the human side of the machine

Ten days into this series, one theme keeps resurfacing: AI systems inherit everything from their data — including our flaws.

- ◆ Consent — was the data the model was trained on used with permission?
- ◆ Transparency — do users know they're interacting with AI, and how decisions get made?
- ◆ Accountability — when an AI system causes harm, who is responsible: the developer, the deployer, or the user?

DAY 10 / 30

“Which AI ethics issue concerns you most: bias, privacy, or accountability?”

Follow along – Day 11 drops tomorrow.

#AIEthics

#ResponsibleAI

#MachineLearning

#AI